

Exercise 1

Consider the two-stage amplifier in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{DD} = 9\text{ V}$, $R_g = 1\text{ k}\Omega$, $R_{G1} = R_{G2} = 100\text{ k}\Omega$, $R_D = 5.6\text{ k}\Omega$, $R_S = 1\text{ k}\Omega$, $R_C = 10\text{ k}\Omega$, $R_E = 2.2\text{ k}\Omega$;
 $R_F = 20\text{ k}\Omega$, $C_g = 10\text{ nF}$, $C_F = 1\text{ }\mu\text{F}$.

Q_1 : $V_{BE} = -0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

Q_2 : $V_t = 3\text{ V}$; $I_{DSS} = 2.25\text{ mA}$; $r_0 = +\infty$;

$V_T = 25\text{ mV}$.

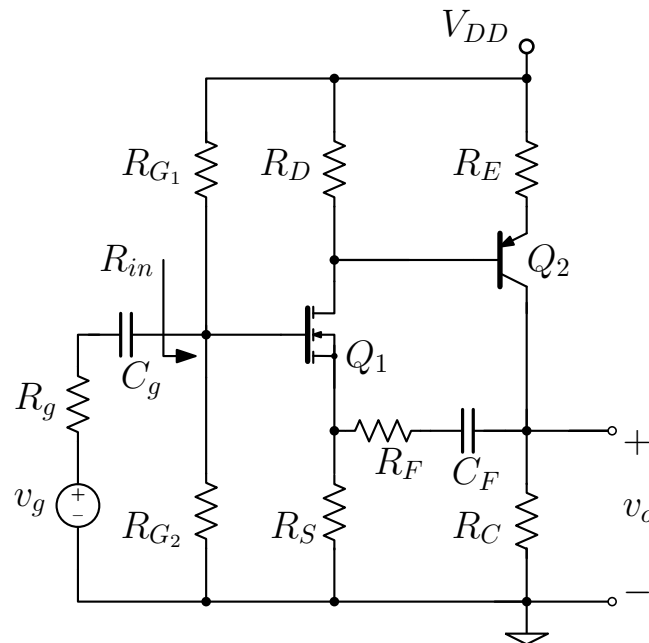


Fig. 1 – Two stage amplifier with feedback.

Exercise 2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 1 \text{ k}\Omega$; $R_2 = 33 \text{ k}\Omega$.

$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)} \text{ [V/V]}, \text{ with } A_0 = 10^5 \text{ [V/V]}; \omega_{p1} = 10^2 \text{ rad/s}; \omega_{p2} = 5 \cdot 10^3 \text{ rad/s}.$$

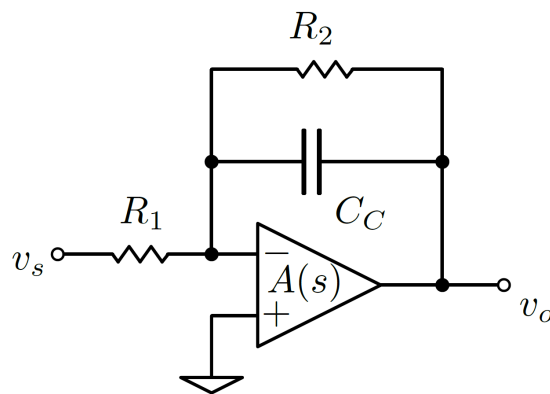


Fig. 2 – Operational amplifier circuit with external compensation.